Measure, optimize, measure again

Personal homepage and Memory Match team project

Personal homepage: 87 → 94 / 100 Performance

LCP: 3.8 s → 2.9 s | Total blocking time: 0 ms → 0 ms
CLS: 0.004 → 0.004 | Transfer: 236 → 197 KiB

Memory Match: 99 → 100 / 100 Performance

LCP: 1.7 s → 1.5 s | Total blocking time: 0 ms → 0 ms
CLS: 0.018 → 0.018 | Transfer: 1276 → 98 KiB

Method and evidence

Lighthouse 13.6.0; Chrome 156; mobile emulation (Moto G Power); simulated throttling: 150 ms RTT, 1,638.4 Kbps throughput and 4x CPU slowdown. One navigation audit per page before and after, each using Lighthouse's cleared-storage default. Both versions were served at http://127.0.0.1:4173/ on the same computer. These are local lab measurements; published-site and real-user results can differ, and scores vary between runs.

Starting issues
Homepage: nested font CSS imports, render-blocking styles (estimated 1,180 ms savings), repeated CSS/JS requests, and a 4.33:1 contrast failure. Memory Match: font import chain, unminified game code, and an off-screen video that transferred about 1.2 MB without being played. The team project was already above the required score of 80.

Scope
Personal site: homepage. Team project: teamproject/index.html. Source baseline: bbe14834d03d70853ddf6440e7997f3dd6cb0046. No scores were invented; pages 3-6 are screenshots of the saved Lighthouse HTML reports.

Changes and reflection

At least five optimizations, with code evidence

1. Discover fonts directly

```
Before: @import url(...) inside the stylesheet.
After: <link rel="stylesheet" href="https://fonts.googleapis.com/css2?...&display=swap">
```

Moves font stylesheet discovery into the HTML instead of waiting for local CSS.

2. Establish font connections early

```
Before: no connection hints.
After: <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
```

Also preconnect to fonts.googleapis.com; starts DNS/TLS earlier.

3. Bundle and minify CSS

```
Before: styles.css + premium.css + project-art.css.
After: <link rel="stylesheet" href="assets/optimized/home.min.css">
```

One minified homepage CSS file; game.min.css serves the team page.

4. Bundle and minify JavaScript

```
Before: game.mjs imports engine.mjs; homepage has an import graph.
After: <script type="module" src="../assets/optimized/game.min.mjs"></script>
```

esbuild bundles modules, preserves module deferral and reduces bytes/request chains.

5. Load the presentation only on demand

```
Before: <video controls preload="metadata">
After: <video controls preload="none">
```

Keeps the off-screen recording available without its initial metadata download.

6. Schedule decoration after primary content

```
Before: companion script starts during initial page setup.
After: addEventListener("load", () => requestIdleCallback(load, {timeout:1500}))
```

Includes a setTimeout fallback; game/navigation retain loading priority.

Reflection

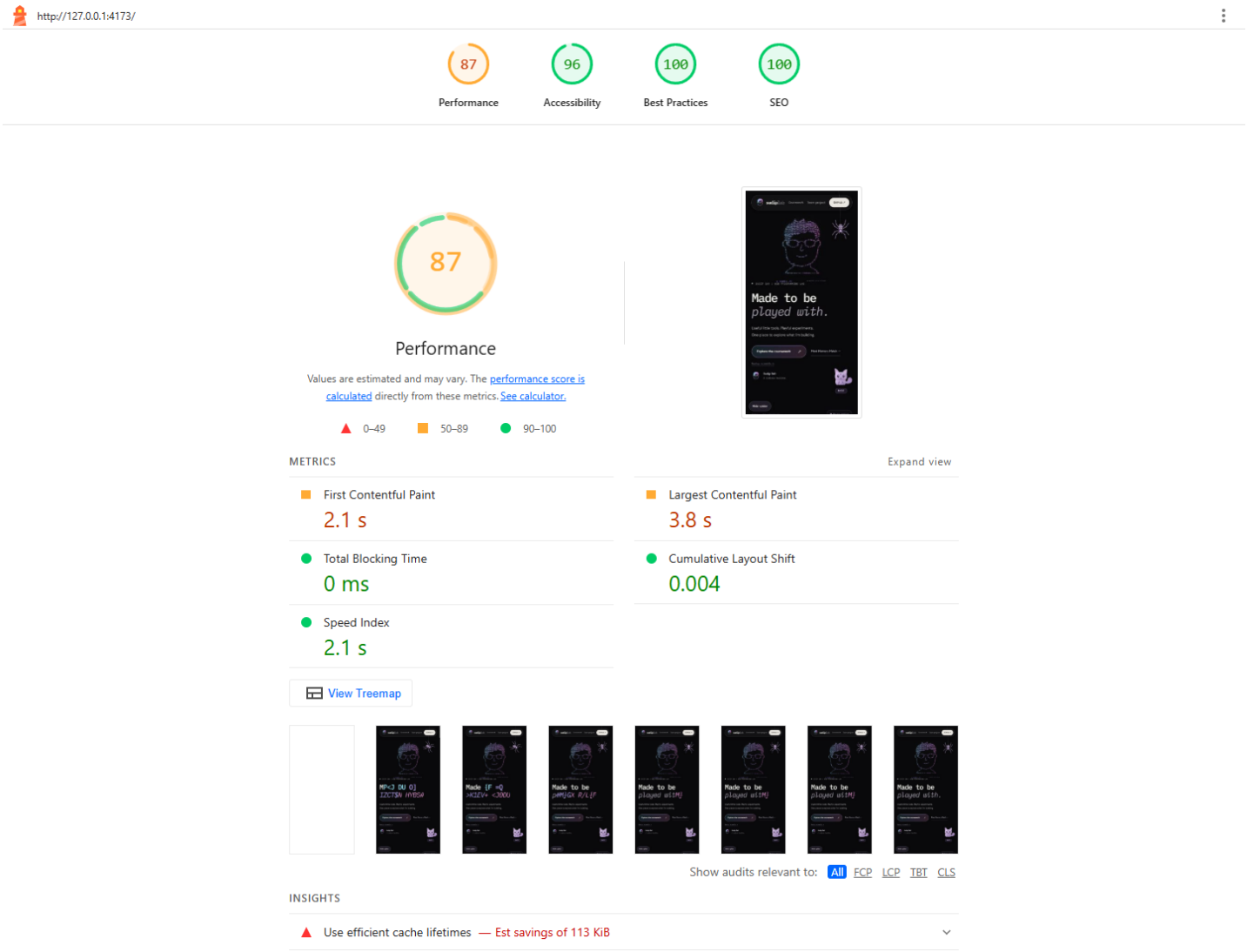
Font discovery and asset bundling were the most relevant changes for the homepage: LCP fell from 3.8 s to 2.9 s in the combined after run. For Memory Match, on-demand video loading produced the clearest bandwidth saving; the total measured transfer fell from 1,307,122 to 99,913 bytes. These combined audits do not isolate the effect of each change.

Other fixes and validation

Homepage caption contrast improved; Accessibility rose from 96 to 100. Added the game meta description. Local link/syntax checker: 0 errors; 11 existing tests passed. Browser checks passed for filtering, idea shuffle, companion loading, practice mode, card flip, video preload and mobile layout. The site uses SVG artwork with image dimensions already set; unnecessary JPEG-to-WebP conversion and hero lazy loading were avoided.

Personal site / before: 87

Original Lighthouse report screenshot - score and metric breakdown

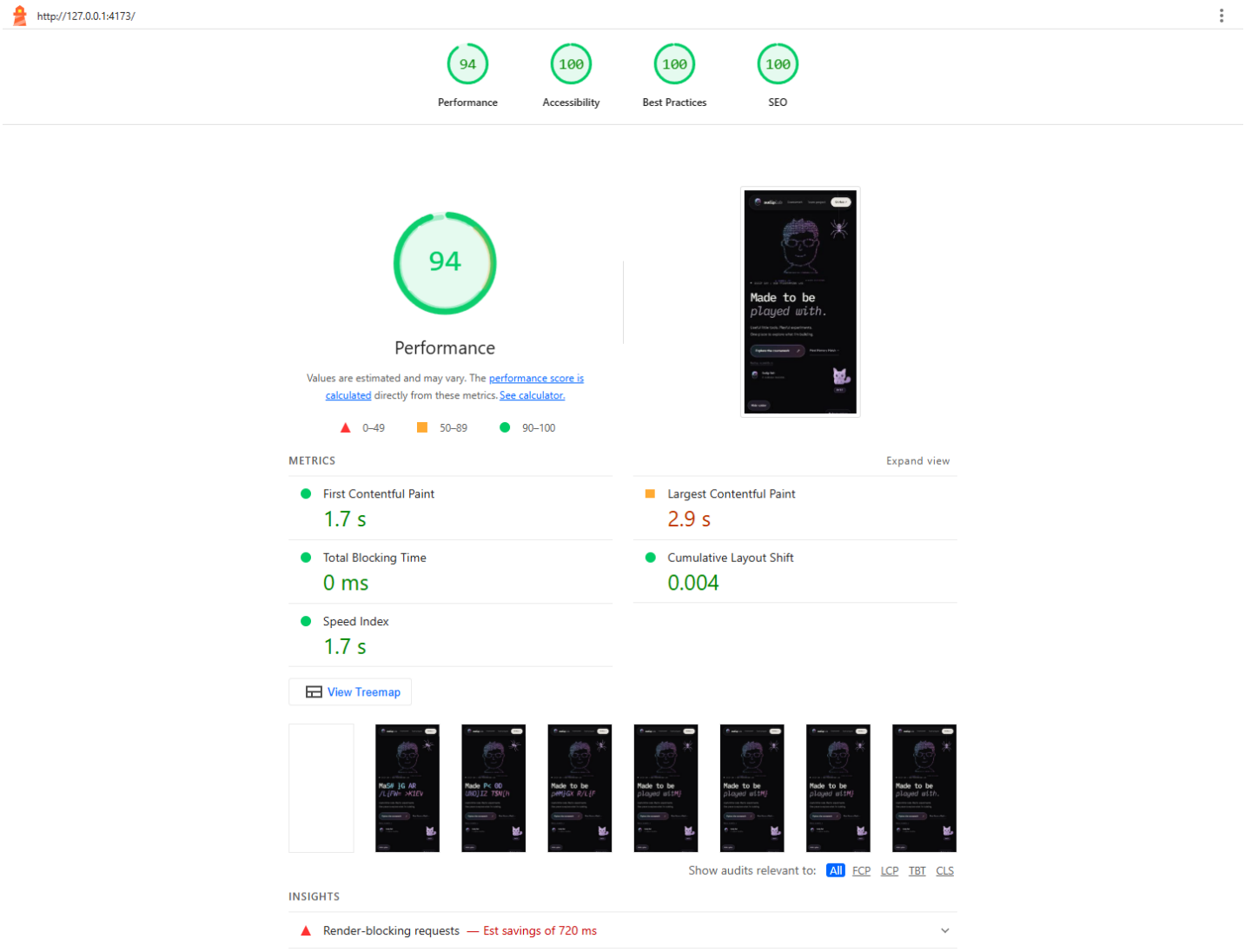


Audit URL: http://127.0.0.1:4173/
Captured: 2026-11-02T14:07:13.742Z (UTC)
Full HTML and JSON reports are supplied alongside this PDF for detailed inspection.

Issues to investigate: Background and foreground colors do not have a sufficient contrast ratio.; Reduce unused CSS; Network dependency tree; Render-blocking requests

Personal site / after: 94

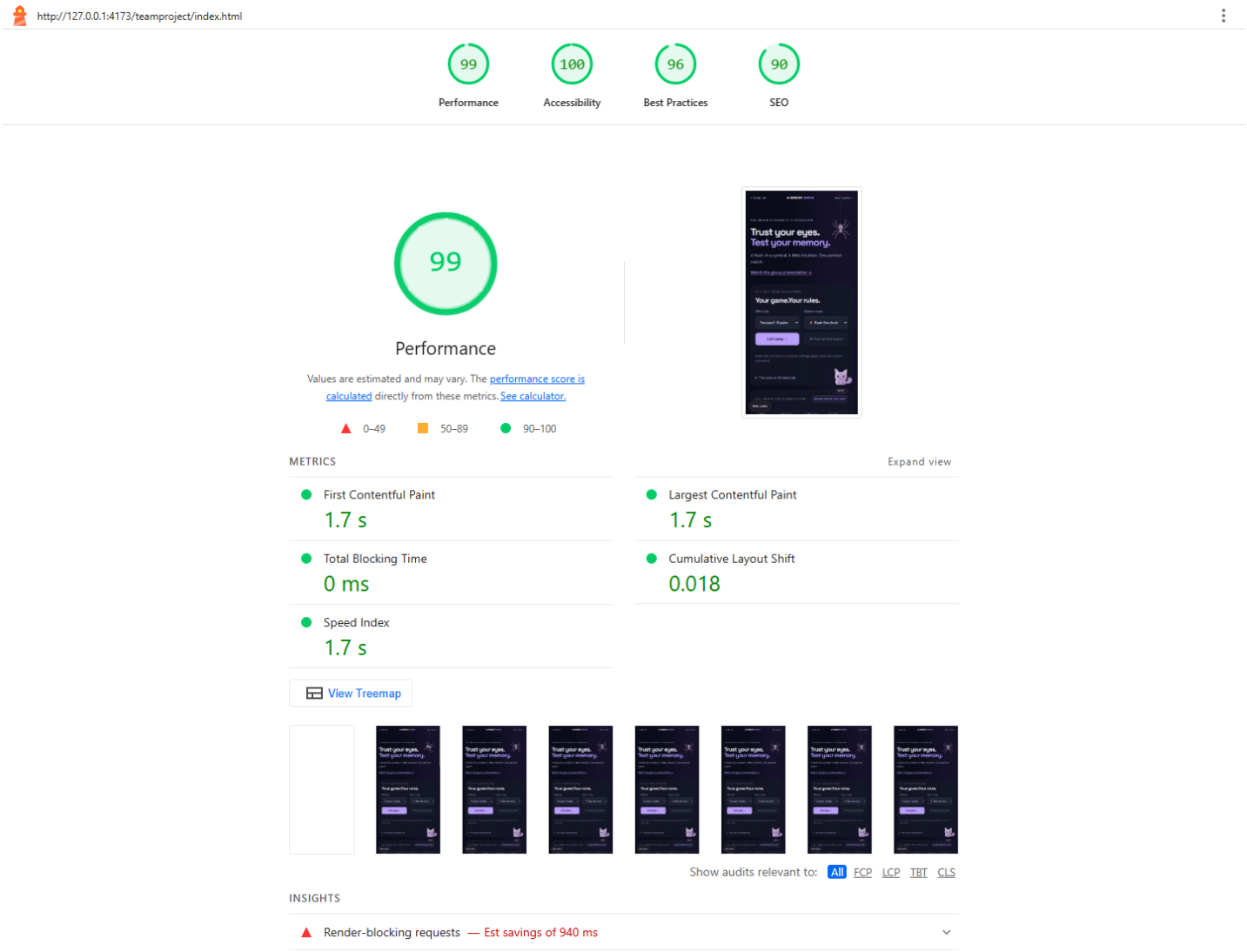
Original Lighthouse report screenshot - score and metric breakdown



Audit URL: http://127.0.0.1:4173/
Captured: 2026-11-02T14:12:43.301Z (UTC)
Full HTML and JSON reports are supplied alongside this PDF for detailed inspection.

Memory Match / before: 99

Original Lighthouse report screenshot - score and metric breakdown

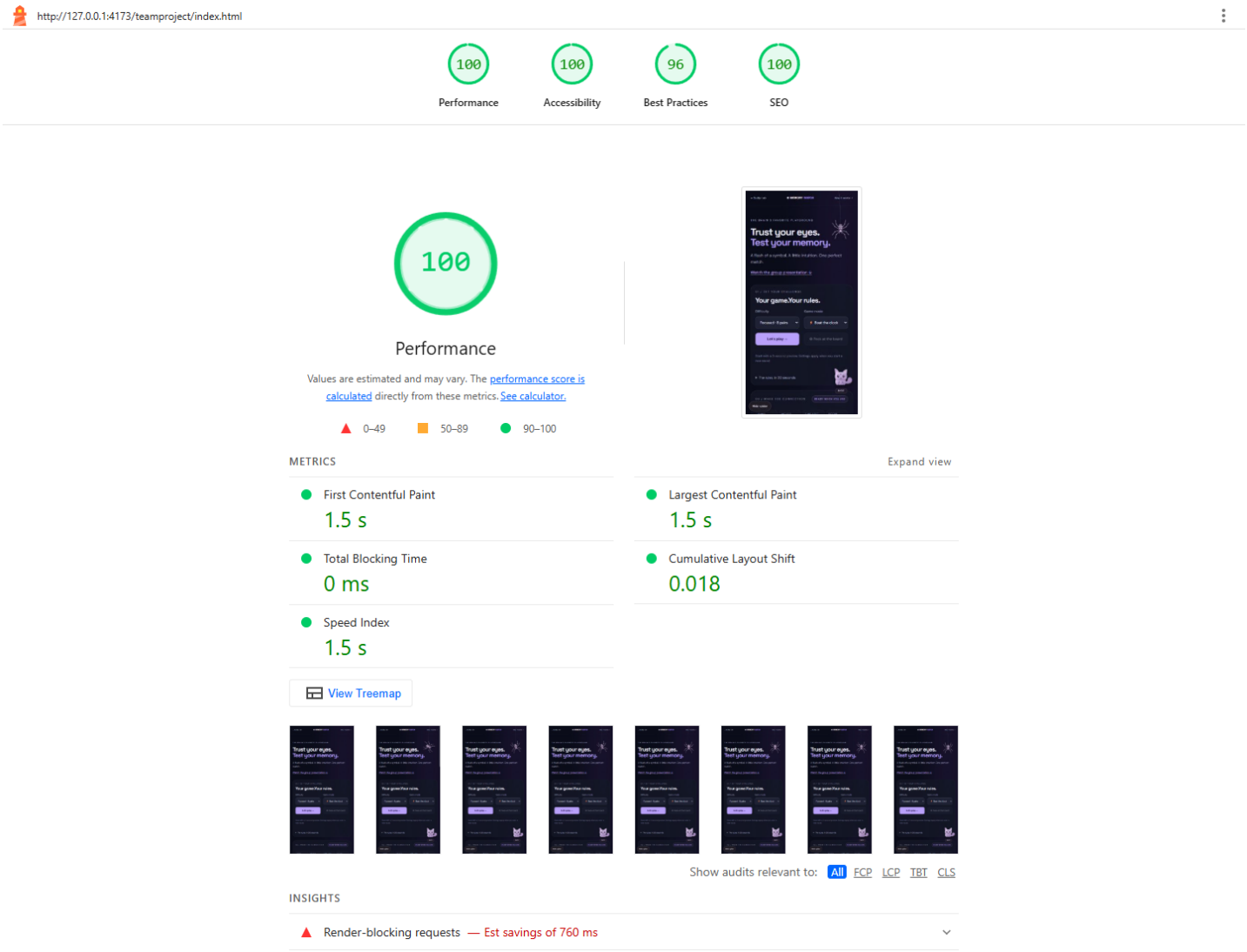


Audit URL: http://127.0.0.1:4173/teamproject/index.html
Captured: 2026-11-02T14:07:45.574Z (UTC)
Full HTML and JSON reports are supplied alongside this PDF for detailed inspection.

Issues to investigate: Network dependency tree; Render-blocking requests

Memory Match / after: 100

Original Lighthouse report screenshot - score and metric breakdown



Audit URL: http://127.0.0.1:4173/teamproject/index.html
Captured: 2026-11-02T14:12:26.537Z (UTC)
Full HTML and JSON reports are supplied alongside this PDF for detailed inspection.